Algorithm 2 Frame Difference Detector (FrameDiffDet)

Require: Downsampled frames $\tilde{f}_t$, $\tilde{f}_{\text{prev}}$
Require: τ_d: pixel difference sensitivity threshold
Ensure: $\mathbf{M}_t \in \{0, 255\}$: binary foreground mask

Convert $\tilde{f}_t$ and $\tilde{f}_{\text{prev}}$ to grayscale: F_t, F_{prev}

2: Compute binary mask, where each pixel (i, j) is set as:

$$\mathbf{M}_t(i,j) \leftarrow \begin{cases} 255 & \text{if } |F_t(i,j) - F_{\text{prev}}(i,j)| > \tau_d \\ 0 & \text{otherwise} \end{cases}$$

return $\mathbf{M}_t$

Algorithm 3 Accumulation Motion Detector (AccMotionDet)

Require: Downsampled frames $\tilde{f}_t$, $\tilde{f}_{\text{prev}}$; accumulated mask $\mathbf{K}_{t-1}$ (initialized to $\mathbf{0}$)

Require: τ_d: pixel difference sensitivity, ω: motion increment, τ_m: activation threshold, $K_{\max}$: accumulation cap, δ: decay rate

Ensure: $\mathbf{M}_t \in \{0, 255\}$: binary foreground mask; $\mathbf{K}_t$: updated accumulated mask

1: Convert $\tilde{f}_t$ and $\tilde{f}_{\text{prev}}$ to grayscale: F_t, F_{prev}

2: $\mathbf{D}_t \leftarrow \left| F_t - F_{\text{prev}} \right|$ ▷ Absolute inter-frame difference

3: $\mathbf{\Delta}_t(i,j) \leftarrow \begin{cases} \omega & \mathbf{D}_t(i,j) > \tau_d \\ 0 & \text{otherwise} \end{cases}$ ▷ Instantaneous increment map

4: $\mathbf{K}_t \leftarrow \min(\mathbf{K}_{t-1} + \mathbf{\Delta}_t,\ K_{\max})$ ▷ Accumulate and cap

5: $\mathbf{K}_t \leftarrow \max(\mathbf{K}_t - \delta,\ 0)$ ▷ Temporal decay

6: $\mathbf{M}_t(i,j) \leftarrow \begin{cases} 255 & \mathbf{K}_t(i,j) \geq \tau_m \\ 0 & \text{otherwise} \end{cases}$ ▷ Threshold to binary mask

7: **return** $\mathbf{M}_t$, $\mathbf{K}_t$

Algorithm 4 Skip Module Hyperparameter Selection

Require: Parameter space Θ, validation set D_{val}, DL model $\mathcal{M}$, recall tolerance δ_R, weights w_S, w_R
Ensure: Optimal parameters θ^*

baseline $\leftarrow$ EVALUATEPIPELINE(D_{val}, skip=None, $\mathcal{M}$)
Extract R_{base} from baseline
$\Theta_{\text{feasible}} \leftarrow \emptyset$
$\triangleright$ **Pass 1: Collect feasible candidates**
for each $\theta \in \Theta$ **do**
    results $\leftarrow$ EVALUATEPIPELINE(D_{val}, $\mathcal{S}(\theta)$, $\mathcal{M}$)
    Extract $R(\theta)$, $S_r(\theta)$ from results
    if $R(\theta) \geq R_{\text{base}} - \delta_R$ **then**
        $\rho(\theta) \leftarrow 1 - \dfrac{R_{\text{base}} - R(\theta)}{\delta_R}$
        $\Theta_{\text{feasible}} \leftarrow \Theta_{\text{feasible}} \cup \{(\theta,\ \rho(\theta),\ S_r(\theta))\}$
    end if
end for
$\triangleright$ **Pass 2: Range-normalize and select**
$\tilde{\rho}(\theta) \leftarrow \dfrac{\rho(\theta) - \min_{\Theta_{\text{feasible}}} \rho}{\max_{\Theta_{\text{feasible}}} \rho - \min_{\Theta_{\text{feasible}}} \rho}$ for each $\theta \in \Theta_{\text{feasible}}$
$\tilde{S}_r(\theta) \leftarrow \dfrac{S_r(\theta) - \min_{\Theta_{\text{feasible}}} S_r}{\max_{\Theta_{\text{feasible}}} S_r - \min_{\Theta_{\text{feasible}}} S_r}$ for each $\theta \in \Theta_{\text{feasible}}$
$\theta^* \leftarrow \arg\max_{\theta \in \Theta_{\text{feasible}}} \left[w_R\, \tilde{\rho}(\theta) + w_S\, \tilde{S}_r(\theta) \right]$
return θ^*

Algorithm 1 Skip Module $\mathcal{S}$: Motion-Based Frame Gating

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Require: Video stream $\{f_1, f_2, ...\}$, motion detector $\mathcal{D}$, scale factor α, block size B, block active threshold τ

Ensure: Skip decisions $\{s_1, s_2, ...\}$, where $s_t \in \{0, 1\}$

$\tilde{f}_{\text{prev}} \leftarrow$ null ▷ Initialize previous downsampled frame
for each incoming frame f_t **do**
    Step 1: Pre-process
    Downsample f_t by α to obtain $\tilde{f}_t$
    Let $B_s \leftarrow B \cdot \alpha$ ▷ Effective block size on downsampled frame
    Pad $\tilde{f}_t$ to be divisible by B_s
    Step 2: Compute Foreground Mask
    if $\tilde{f}_{\text{prev}} =$ null **then** ▷ No previous frame on first iteration
        $s_t \leftarrow 1$ ▷ Run inference on first frame
        $\tilde{f}_{\text{prev}} \leftarrow \tilde{f}_t$
        continue
    end if
    $\mathbf{M}_t \leftarrow \mathcal{D}(\tilde{f}_t, \tilde{f}_{\text{prev}})$ ▷ $\mathcal{D}$: e.g., FrameDiffDet or AccMotionDet
    Partition $\mathbf{M}_t$ into a grid of non-overlapping blocks of size B_s
    Mark a block as *active* if its foreground pixel ratio $> \tau$
    Step 3: Gating Decision
    if any active block exists **then**
        $s_t \leftarrow 1$ ▷ Motion detected, run DL inference
    else
        $s_t \leftarrow 0$ ▷ No motion, skip inference
    end if
    $\tilde{f}_{\text{prev}} \leftarrow \tilde{f}_t$ ▷ Update previous frame for next iteration
    yield s_t
end for
